



Hospital Emergency Patient Management System

Severity-Based Treatment & FIFO Arrival Processing Engine

C Max-Heap Backend Active

Seed Demo Cases

+25 Heap Stress Test

Reset All

Total Waiting Patients

5

Critical Patients (Level 5)

2

Patients Treated

0

Currently Being Treated

No active treatment

PRIORITY QUEUE ROOT

Next Up: ID #102 - Clara Oswald (P102)

Severity: Critical (Level 5) | Arrival Order: #2 | Assigned: Dr. Alexander Ross

Treat Next Highest-Priority Patient

Register New Patient

Automatically inserted into C Max-Heap

Patient Full Name *

e.g. Eleanor Vance

Triage Severity Level *

Level 5 - Critical (Immediate Resuscita

Assigned Doctor

e.g. Dr. Sarah Jenkins

Waiting Patients Priority Queue

Ordered strictly by Max Severity, then FIFO Arrival Number

5 Patients

PATIENT ID	PATIENT NAME	ARRIVAL NO.	SEVERITY	PRIORITY RANK	ASSIGNED DOCTOR	STATUS	ACTIONS
#102	Clara Oswald (P102)	Arrival #2	5 - CRITICAL	#1 ROOT (TOP)	Dr. Alexander Ross	Waiting	Edit Delete
#104	Elena Rostova (P104)	Arrival #4	5 - CRITICAL	#2	Dr. Marcus Vance	Waiting	Edit Delete



C Backend Max-Heap Memory Inspection Visualizer

Direct array index mapping of the binary heap in C RAM (Parent: (i-1)/2, Left: 2i+1, Right: 2i+2)

Refresh Visualizer

[Index 0] * ROOT
#104 Elena Rostova (P104)
5 - CRITICAL
Arr: #4
Parent: None
Left: [1] | Right: [2]

[Index 1]
#101 Arthur Pendelton (P101)
2 - MILD
Arr: #1
Parent: [0]
Left: [3] | Right: None

[Index 2]
#103 David Zhang (P103)
3 - MODERATE
Arr: #3
Parent: [0]
Left: None | Right: None

[Index 3]
#105 Felix Vance (P105)
2 - MILD
Arr: #5
Parent: [1]
Left: None | Right: None

Treatment History Log

Completed patient treatments dispatched from Max Heap

1 Treated

PATIENT ID	PATIENT NAME	ARRIVAL NO.	SEVERITY	DOCTOR	ARRIVAL TIME	DISPATCHED TIME	STATUS
#182	Clara Oswald (P102)	Arrival #2	5 - CRITICAL	Dr. Alexander Ross	14:53:46	14:54:07	Treated

Dispatched Patient #102 (Clara Oswald (P102)) for treatment!

Register New Patient

Automatically inserted into C Max-Heap

Patient Full Name *

Mani

Triage Severity Level *

Level 2 - Mild (Semi-urgent)▼

Assigned Doctor

Dr. Joseph

+ Insert Patient into Priority Queue

Q Search Patient by ID

Enter Patient ID (e.g. 101)

Search

Waiting Patients Priority

Ordered strictly by Max Severity, then P

PATIENT ID	PATIENT NAME
#104	Elena Rostova (P104)
#103	David Zhang (P103)
#101	Arthur Pendelton (P101)
#105	Felix Vance (P105)

Q Search Patient by ID

106

Search

Patient #106: Mani

2 - MILD

Arrival Order: #6 | Doctor: Dr. Joseph

Arrival Time: 14:54:53

Current Status: Waiting (Heap Index 4)

#105	Felix Vance (P105)	Arrival #5	2 - MILD	#4	Dr. Sarah Jenkins	Waiting	EditDelete
#106	Mani	Arrival #6	2 - MILD	#5	Dr. Joseph	Waiting	EditDelete

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Refresh Visualizer